

scenario_A_reproduce

June 3, 2026

```
[47]: %load_ext autotime

# 0. Imports & path setup
import sys, os, copy, time, warnings
# sys.path.insert(0, "/home/niels/FELsim_clone/backend")
CURRENT_DIR = os.getcwd()
BACKEND_DIR = CURRENT_DIR if os.path.basename(CURRENT_DIR) == "backend" else os.
↳path.abspath(os.path.join(CURRENT_DIR, "../backend"))
sys.path.insert(0, BACKEND_DIR)
print(BACKEND_DIR)
os.makedirs(os.path.join(CURRENT_DIR, "../results"), exist_ok=True)

import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("module://matplotlib_inline.backend_inline")
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from matplotlib.lines import Line2D

from ebeam import beam as ebeam_class
from beamline import lattice
from excelElements import ExcelElements
from beamOptimizer import beamOptimizer
from evolutionPlotter import EvolutionPlotter
from felsimAdapter import FELsimAdapter

# Shared constants
# EXCEL_PATH      = "/home/niels/FELsim_clone/beam_excel/Beamline_elements_3.
↳xlsx"
# EXCEL_PATH      = "/home/niels/FELsim_clone/beam_excel/Beamline_elements_3.
↳xlsx"
EXCEL_PATH = os.path.abspath(os.path.join(CURRENT_DIR, "../beam_excel/
↳Beamline_elements_3.xlsx"))

BEAM_ENERGY      = 40.0      # MeV
```

```

N_PARTICLES      = 1000
SEED              = 42
CURRENT_BOUNDS   = (0.01, 1.5)    # A (lower bound > 0)

# Relativistic factors
relat = lattice(1, fringeType=None)
relat.setE(E=BEAM_ENERGY)
GAMMA_REL = relat.gamma
BETA_REL  = relat.beta
NORM      = GAMMA_REL * BETA_REL

# Undulator matching targets (MkV FEL, K=1.2, _u=2.3 cm)
K          = 1.2
LAMBDA_U   = 2.3e-2    # m
BETA_YM    = GAMMA_REL / (K * 2 * np.pi / LAMBDA_U)    # matched _y at UND_
↳ entrance
BETA_XM    = 1.4        # m - matched _x at UND entrance
ALPHA_XM   = 0.47       # - Courant-Snyder _x at UND entrance
ALPHA_YM   = 0.0        # - Courant-Snyder _y at UND entrance

print(f" = {GAMMA_REL:.4f},    = {BETA_REL:.6f},    = {NORM:.4f}")
print(f"Matched _ym = {BETA_YM:.4f} m    (K={K}, _u={LAMBDA_U*100:.1f} cm)")
print(f"Matched _xm = {BETA_XM:.4f} m,    _xm = {ALPHA_XM:.4f},    _ym =_
↳ {ALPHA_YM:.4f}")

```

The autotime extension is already loaded. To reload it, use:

```

%reload_ext autotime
c:\Users\yilu2\my_files\research\FELsim\backend
= 79.2794,    = 0.999920,    = 79.2731
Matched _ym = 0.2418 m    (K=1.2, _u=2.3 cm)
Matched _xm = 1.4000 m,    _xm = 0.4700,    _ym = 0.0000
time: 0 ns (started: 2026-06-02 16:39:07 -10:00)

```

```

[48]: import warnings

warnings.filterwarnings(
    "ignore",
    message=".*'where' used without 'out'.*",
    category=UserWarning,
    module="scipy.optimize._trustregion_constr"
)

warnings.filterwarnings(
    "ignore",
    message=".*delta_grad == 0.0.*",
    category=UserWarning,
    module="scipy.optimize._differentiable_functions"
)

```

```
)
```

time: 0 ns (started: 2026-06-02 16:39:08 -10:00)

0.1 1 — Beam Distribution

Physics-based 6D Gaussian matching the UH linac reference parameters.

```
[49]: %load_ext autotime

# Beam parameters
epsilon_n      = 8.0      # .mm·mrad  normalised emittance
x_std          = 0.8      # mm
y_std          = 0.8      # mm
f_RF           = 2856e6   # Hz
bunch_spread_ps = 2.0     # ps
energy_spread_pct = 0.5   # %
h              = 5e9      # 1/s  chirp

epsilon        = epsilon_n / NORM
x_prime_std    = epsilon / x_std
y_prime_std    = epsilon / y_std
tof_std        = bunch_spread_ps * 1e-9 * f_RF
energy_std     = energy_spread_pct * 10

ebeam_obj = ebeam_class()
PARTICLES = ebeam_obj.gen_6d_gaussian(
    0,
    [x_std, x_prime_std, y_std, y_prime_std, tof_std, energy_std],
    N_PARTICLES
)

# Apply longitudinal chirp
PARTICLES[:, 5] += h * (PARTICLES[:, 4] / f_RF)

print(f" = {epsilon:.5f} .mm·mrad (geometric)")
print(f"_x={x_std:.2f} mm, _x'={x_prime_std:.5f} mrad")
print(f"_y={y_std:.2f} mm, _y'={y_prime_std:.5f} mrad")
print(f"Particles: {PARTICLES.shape}")
```

The autotime extension is already loaded. To reload it, use:

```
%reload_ext autotime
= 0.10092 .mm·mrad (geometric)
_x=0.80 mm, _x'=0.12615 mrad
_y=0.80 mm, _y'=0.12615 mrad
Particles: torch.Size([1000, 6])
time: 0 ns (started: 2026-06-02 16:39:08 -10:00)
```

0.2 2 — Beamline & Element Index Map

The `ExcelElements` parser produces **138 FELsim elements**. Key reference indices: | Location | FELsim idx | z (m) | Physical meaning | |—|—|—|—| | End of 1st doublet region | 8–9 | 0.73–1.0 | First targets | | IP (focus) | 59 | 7.11 | Minimum envelope | | UND entrance | 117 | 12.39 | Undulator matching |

```
[50]: %load_ext autotime

# Load beamline & map quad indices
beamline_full = ExcelElements(EXCEL_PATH).create_beamline()
print(f"FELsim elements: {len(beamline_full)}")

s = 0.0
quad_info = []
for i, e in enumerate(beamline_full):
    cls = e.__class__.__name__
    L = float(getattr(e, 'length', 0.0) or 0.0)
    if cls in ('qpfLattice', 'qpdLattice'):
        quad_info.append((i, cls, round(s, 5), round(s+L, 5),
                          round(float(getattr(e, 'current', 0) or 0), 5)))
    s += L

quad_indices = [q[0] for q in quad_info]
df_quads = pd.DataFrame(quad_info,
    columns=['idx', 'type', 's_start', 's_end', 'current_A'])
print(f"Quadrupoles: {len(quad_indices)}")
print(df_quads.to_string(index=False))

# Gold-standard currents from Eremey's FELsim S1 solution
FELSIM_S1_CURRENTS = {
    1: 0.8218, 3: 1.0430,
    10: 3.8834,
    16: 2.2396, 18: 4.9532, 20: 3.4258,
    27: 4.6657,
    33: 2.6942, 35: 2.6523, 37: 0.2768, 39: 0.2768, 41: 2.6523, 43: 2.6942,
    50: 4.6739,
    56: 3.1219, 58: 3.3129,
    61: 5.1775, 63: 4.0434,
    70: 4.6818,
    76: 3.9336, 78: 4.0787, 80: 0.0139,
    87: 1.3624, 93: 0.9452, 95: 2.8851, 97: 2.1921,
}
print(f"\nFELSIM_S1_CURRENTS: {len(FELSIM_S1_CURRENTS)} quadrupoles loaded")
```

The autotime extension is already loaded. To reload it, use:

```
%reload_ext autotime
```

FELsim elements: 138

Quadrupoles: 26

idx	type	s_start	s_end	current_A
1	qpfLattice	0.35878	0.44768	0.88572
3	qpdLattice	0.64453	0.73342	1.05289
10	qpfLattice	2.03304	2.12194	3.81660
16	qpdLattice	2.73535	2.82425	2.57981
18	qpfLattice	2.86743	2.95633	4.56570
20	qpdLattice	2.99951	3.08841	2.57981
27	qpfLattice	3.55448	3.64338	4.49146
33	qpdLattice	4.15187	4.24077	1.59614
35	qpfLattice	4.29157	4.38047	3.08092
37	qpdLattice	4.43127	4.52017	1.59614
39	qpdLattice	4.93513	5.02403	1.59614
41	qpfLattice	5.07483	5.16373	3.08092
43	qpdLattice	5.21453	5.30343	1.59614
50	qpfLattice	5.83098	5.91988	4.49072
56	qpdLattice	6.36946	6.45836	2.25575
58	qpfLattice	6.55775	6.64665	1.67186
61	qpfLattice	7.32562	7.41452	0.02487
63	qpdLattice	7.46992	7.55882	0.00100
70	qpfLattice	8.10749	8.19639	4.49146
76	qpdLattice	8.60769	8.69659	1.59614
78	qpfLattice	8.73278	8.82168	3.50000
80	qpdLattice	8.85788	8.94678	1.59614
87	qpfLattice	9.62889	9.71779	3.81660
93	qpdLattice	10.21208	10.30098	1.59614
95	qpfLattice	10.60921	10.69811	3.08092
97	qpdLattice	10.73621	10.82511	1.59614

FELSIM_S1_CURRENTS: 26 quadrupoles loaded
time: 141 ms (started: 2026-06-02 16:39:08 -10:00)

```
[51]: %load_ext autotime

# Confirm key reference positions
s = 0.0
for i, e in enumerate(beamline_full):
    s += float(getattr(e, 'length', 0.0) or 0.0)
    if i in (8, 9, 15, 25, 32, 37, 55, 59, 92, 117):
        print(f" el {i:3d} {e.__class__.__name__:20s} s_end={s:.5f} m")
```

The autotime extension is already loaded. To reload it, use:

```
%reload_ext autotime
el  8  dipole_wedge          s_end=1.80682 m
el  9  driftLattice          s_end=2.03304 m
el 15  driftLattice          s_end=2.73535 m
el 25  dipole_wedge          s_end=3.34855 m
el 32  driftLattice          s_end=4.15187 m
```

```

el 37 qpdLattice          s_end=4.52017 m
el 55 driftLattice        s_end=6.36946 m
el 59 driftLattice        s_end=7.11013 m
el 92 driftLattice        s_end=10.21208 m
el 117 driftLattice       s_end=12.39158 m
time: 0 ns (started: 2026-06-02 16:39:08 -10:00)

```

0.3 3 — Statistical Benchmark Infrastructure

`run_benchmark()` runs the same scenario `N_RUNS` times from different random starting currents. `plot_stat_convergence()` plots **mean $\pm$ 1 RMS bands** — the same visual style as Eremey's reference convergence figures.

```

[52]: %load_ext autotime

import time
import copy
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.lines import Line2D

def method_label(method, label=None):
    if label:
        return label
    if isinstance(method, str):
        return method
    return getattr(method, "__name__", str(method))

# Benchmark runner
def run_benchmark(scenario_name, beamline_slice_len, particles, seg_var, obj,
    bounds,
                    method, n_runs=5, SEED=42, seed_offset=0, jac=None,
    options=None, verbose=True, method_tag=None):
    """Run beamOptimizer.calc() n_runs times from uniform-random starting
    currents."""
    var_names = list({seg_var[i][0] for i in seg_var})
    results = []
    method_name = method_label(method, method_tag)

    for run_i in range(n_runs):
        bl = ExcelElements(EXCEL_PATH).create_beamline()[0:beamline_slice_len]
        p = particles.clone()
        rng = np.random.default_rng(SEED + seed_offset + run_i * 997)
        sp = {v: {"bounds": bounds[v],
                  "start": float(rng.uniform(bounds[v][0], bounds[v][1]))}
              for v in var_names}
        start_x = {v: sp[v]["start"] for v in var_names}

```

```

obj_copy = copy.deepcopy(obj)
opti = beamOptimizer(bl, p)
t0 = time.perf_counter()

try:
    res = opti.calc(method, seg_var, sp, obj_copy, jac=jac,
↳options=options,
                                printResults=False, plotProgress=False)
    wall = time.perf_counter() - t0

    result_x_dict = dict(zip(var_names, res.x))
    i_1 = result_x_dict.get("I", np.nan)
    i_3 = result_x_dict.get("I2", np.nan)

    def safe_get_measured(idx):
        if idx in opti.objectives and len(opti.objectives[idx]) > 0:
            val = opti.objectives[idx][0].get("measured", np.nan)
            return float(val.item() if hasattr(val, 'item') else val)
        return np.nan

    alpha_x = safe_get_measured(8)
    alpha_y = safe_get_measured(9)

    results.append({
        "run": run_i+1, "method": method_name, "scenario":
↳scenario_name,
        "final_mse": float(res.fun),
        "nfev": int(getattr(res, 'nfev', len(opti.plotMSE))),
        "nit": int(getattr(res, 'nit', -1)),
        "wall_time": round(wall, 3),
        "mse_curve": list(opti.plotMSE), "success": bool(res.success),
        "start_x": start_x, "result_x": result_x_dict,
        "I_1": i_1, "I_3": i_3, "alpha_x": alpha_x, "alpha_y": alpha_y
    })
except Exception as exc:
    wall = time.perf_counter() - t0
    results.append({
        "run": run_i+1, "method": method_name, "scenario":
↳scenario_name,
        "final_mse": float("nan"), "nfev": -1, "nit": -1,
        "wall_time": round(wall, 3), "mse_curve": [], "success": False,
        "start_x": start_x, "result_x": {}, "error": str(exc),
        "I_1": np.nan, "I_3": np.nan, "alpha_x": np.nan, "alpha_y": np.
↳nan
    })
    if verbose: print(f" {method_name} run {run_i+1} FAILED: {exc}")
    continue

```

```

        if verbose:
            print(f" {method_name:22s} run {run_i+1}/{n_runs} "
                  f"MSE={results[-1]['final_mse']:.3e} nit={results[-1]['nit']}:
↳<3} "
                  f"nfev={results[-1]['nfev']:<4} t={wall:.2f}s")
        return results

def results_to_df(results_list):
    keep = ('run', 'method', 'scenario', 'method_tag', 'final_mse', 'nfev', 'nit',
            'wall_time', 'success', 'I_1', 'I_3', 'alpha_x', 'alpha_y')
    return pd.DataFrame([k: r[k] for k in keep if k in r for r in
↳results_list])

def plot_stat_convergence(results_by_tag, title="Convergence", figsize=None,
                           colors=None, ax=None, save_path=None, scale="log",
                           convergence_epsilon=None):
    standalone = ax is None
    if standalone: fig, ax = plt.subplots(figsize=figsize or (7, 4.5))
    else: fig = ax.get_figure()

    # FIX 1: Matplotlib API Update
    try:
        cmap = plt.colormaps.get_cmap("tab10")
    except AttributeError:
        cmap = plt.get_cmap("tab10")

    if colors is None: colors = [cmap(i) for i in range(len(results_by_tag))]

    legend_handles = []
    for (tag, res_list), col in zip(results_by_tag.items(), colors):
        n_t = len(res_list)

        if convergence_epsilon is None:
            converged_iter = [r for r in res_list if r.get('mse_curve')]
        else:
            converged_iter = [r for r in res_list
                              if r.get('mse_curve')
                              and np.isfinite(r['mse_curve'][-1])
                              and r['mse_curve'][-1] < convergence_epsilon]

        n_s = len(converged_iter)
        curves = [r['mse_curve'] for r in converged_iter]

        if not curves:
            legend_handles.append(Line2D([0], [0], color=col, linewidth=2,
                                          label=f"{tag} (0/{n_t} converged)"))
            continue

```



```

max_len = max(len(c) for c in curves)
padded = np.array([c + [c[-1]] * (max_len - len(c)) for c in curves],
dtype=np.float64)

# FIX 2: clip to a positive floor so log10 is well-defined
padded = np.clip(padded, a_min=1e-20, a_max=1e100)

evals = np.arange(1, max_len + 1)

log_padded = np.log10(padded)
log_mean = np.mean(log_padded, axis=0)
log_std = np.std(log_padded, axis=0)
mean = 10 ** log_mean
band_lower = 10 ** (log_mean - log_std)
band_upper = 10 ** (log_mean + log_std)

ax.plot(evals, mean, color=col, linewidth=2.0, zorder=3)
ax.fill_between(evals, np.maximum(band_lower, 1e-20), band_upper,
color=col, alpha=0.20, zorder=2)

legend_handles.append(Line2D([0], [0], color=col, linewidth=2,
label=f"{tag} ({n_s}/{n_t} converged)"))

ax.set_yscale(scale)
ax.set_xlabel('Function evaluations (Forward Passes)', fontsize=11)
ax.set_ylabel('Normalised MSE', fontsize=11)
ax.set_title(title, fontsize=12, fontweight='bold')
ax.legend(handles=legend_handles, fontsize=8, framealpha=0.9, loc='upper_
right')
ax.grid(True, which='both', alpha=0.25)
ax.tick_params(labelsize=9)
if standalone:
    fig.tight_layout()
    if save_path: fig.savefig(save_path, dpi=200, bbox_inches='tight')
    plt.show()
return ax
print("Benchmark helpers loaded (Warnings Suppressed).")

```

The autotime extension is already loaded. To reload it, use:

```
%reload_ext autotime
Benchmark helpers loaded (Warnings Suppressed).
time: 15 ms (started: 2026-06-02 16:39:08 -10:00)
```

0.4 4 — Scenario A: Doublet (2 quadrupoles)

Exact replication of Stage 1 from `beamline_optimization.ipynb`: $x = y = 0$ at elements 8–9 (s 0.73–1.0 m).

Simple 2D problem — validates all methods and compares FELsim vs COSY timing.

```
[53]: def run_scenario_A(CURRENT_BOUNDS, EPSILON, N_RUNS_A, REF_I, A_OBJ,
    ↪A_BEAMLINE_LEN, METHODS_A, SEED=42, scale="log", options=None,
    ↪METHOD_OPTIONS={}):
    A_BOUNDS = {"I1": CURRENT_BOUNDS, "I2": CURRENT_BOUNDS}
    REF_I_1 = REF_I[0]
    REF_I_3 = REF_I[1]
    results_A = {}
    for method_spec in METHODS_A:
        if len(method_spec) == 3:
            method, jac, label = method_spec
        else:
            method, jac = method_spec
            label = None
        options = METHOD_OPTIONS.get(method_label(method, label), None)
        method_name = method_label(method, label)
        tag = method_name + (" +jac" if jac else "")
        print(f"\n Scenario A - {tag} ({N_RUNS_A} runs)")

        # Execute benchmark (Note: passing 'jac' here prepares for PyTorch
    ↪analytical gradients later)
        res = run_benchmark(
            "A",
            A_BEAMLINE_LEN,
            PARTICLES,
            A_VARS,
            A_OBJ,
            A_BOUNDS,
            method=method,
            n_runs=N_RUNS_A,
            SEED=SEED,
            seed_offset=0,
            jac=jac,
            options=options,
            method_tag=method_name,
        )
        for r in res:
            r["method_tag"] = tag
            results_A[tag] = res

    # Flatten and merge nested results into a single DataFrame
    df_A = results_to_df([r for v in results_A.values() for r in v])
```

```

#
↳ =====
# CORE FIX: Patch the bug where derivative-free algorithms return
↳ negative iterations
#
↳ =====
# Since derivative-free algorithms lack internal line searches, their number
# of function evaluations (nfev) is mathematically equivalent to their true
↳ exploration steps (nit).
# If iteration count < 0 is detected, directly extract 'nfev' as the true
↳ iteration count!
if 'nit' in df_A.columns and 'nfev' in df_A.columns:
    df_A.loc[df_A['nit'] < 0, 'nit'] = df_A.loc[df_A['nit'] < 0, 'nfev']

print("\n" + "="*115)
print(f"    SCENARIO A: Robustness (Tol < {EPSILON}) & Discovered Physics
↳ State vs Reference")
print("="*115)

# 1. Core Evaluation: Check if each run successfully converged below the
↳ target MSE threshold
df_A['is_converged'] = df_A['final_mse'] < EPSILON

# 2. Calculate average performance metrics
def _geom_mean_converged(s):
    conv = s[s < EPSILON]
    return 10 ** np.log10(conv).mean() if len(conv) else np.nan
# ( Note: The redundant 'nfev_mean' has been completely removed as
↳ requested)
summary_stats = df_A.groupby('method_tag').agg(
    conv_rate=('is_converged', 'mean'),
    nit_mean=('nit', 'mean'),
    nfev_mean=('nfev', 'mean'),
    wall_time_mean=('wall_time', 'mean'),
).join(
    df_A.groupby('method_tag')['final_mse']
        .apply(_geom_mean_converged)
        .rename('final_mse_mean')
)

# 3. Extract the Champion Solution: Find the run with the lowest error for
↳ each method,
# extracting its exact hardware currents and resulting alpha observables.
idx_best = df_A.groupby('method_tag')['final_mse'].idxmin()

```

```

best_runs = df_A.loc[idx_best, ['method_tag', 'I_1', 'I_3', 'alpha_x',
↪ 'alpha_y']].set_index('method_tag')

# 4. Merge all statistical data
final_summary = summary_stats.join(best_runs).reset_index()

# Inject the expert's gold-standard values to establish the Ground Truth
final_summary['I_1_ref'] = REF_I_1
final_summary['I_3_ref'] = REF_I_3

# Reorder columns: Place the AI-predicted currents right next to the
↪ expert's Reference values
final_summary = final_summary[[
    'method_tag', 'conv_rate', 'nit_mean', 'nfev_mean', 'wall_time_mean',
    'final_mse_mean', 'I_1_ref', 'I_1', 'I_3_ref', 'I_3', 'alpha_x',
↪ 'alpha_y'
]]

# 5. Plain text formatted output to bypass Jupyter HTML rendering bugs
↪ (missing tables)
format_dict = {
    'conv_rate': '{:.0%}',
    'nit_mean': '{:.1f}',
    'nfev_mean': '{:.1f}',
    'wall_time_mean': '{:.3f} s',
    'final_mse_mean': '{:.2e}',
    'I_1_ref': '{:.4f} A',
    'I_1': '{:.4f} A',
    'I_3_ref': '{:.4f} A',
    'I_3': '{:.4f} A',
    'alpha_x': '{:.2e}',
    'alpha_y': '{:.2e}'
}

formatted_df = final_summary.copy()

# Apply formatting conversions in bulk
for col, fmt in format_dict.items():
    if col in formatted_df.columns:
        formatted_df[col] = formatted_df[col].apply(lambda x: fmt.format(x)
↪ if pd.notna(x) else "NaN")

# Print the final beautifully aligned data table
print(formatted_df.to_string(index=False, justify='center'))
print("="*115)

# Plot the final algorithm stability curve

```

```

plot_stat_convergence(
    results_A,
    title=f"Scenario A: Convergence Stability (Target MSE < {EPSILON})",
    convergence_epsilon=EPSILON,
    scale=scale
)
return results_A, df_A, final_summary

```

time: 15 ms (started: 2026-06-02 16:39:08 -10:00)

```

[54]: %load_ext autotime
def display_results(results_A):

    # Scenario A - Best optimised quadrupole currents per method
    print(" Scenario A: Best Optimised Quadrupole Currents per Method ")
    print(f"{'Method':<22} {'Best MSE':>12} {'I (A)':>10} {'I2 (A)':>10}")
    print(" " * 58)
    for tag, res_list in results_A.items():
        valid = [r for r in res_list
                  if r.get('result_x') and not np.isnan(r['final_mse'])]
        if not valid:
            print(f" {tag:<20} no valid run")
            continue
        best = min(valid, key=lambda r: r['final_mse'])
        I = best['result_x'].get('I', float('nan'))
        I2 = best['result_x'].get('I2', float('nan'))
        print(f" {tag:<20} {best['final_mse']:>12.4e} {I:>10.4f} {I2:>10.4f}")
    print()
    print(f" Reference FELsim S1 currents: I={REF_I[0]:.4f} A, I2={REF_I[1]:.4f} A")
    # Mean best across all runs
    all_valid = [r for v in results_A.values() for r in v
                  if r.get('result_x') and not np.isnan(r['final_mse'])]
    if all_valid:
        best_overall = min(all_valid, key=lambda r: r['final_mse'])
        print(f"\n Overall best across all methods/runs:")
        print(f" Method: {best_overall['method_tag']}, I=
MSE={best_overall['final_mse']:.4e}")
        print(f" I={best_overall['result_x']['I']:.5f} A, I2=
f"I2={best_overall['result_x']['I2']:.5f} A")

```

The autotime extension is already loaded. To reload it, use:

```
%reload_ext autotime
```

time: 0 ns (started: 2026-06-02 16:39:08 -10:00)

```

[55]: CURRENT_BOUNDS = (0.01, 1.5)
      EPSILON = 1e-3

```

```

REF_I_1 = 0.8218
REF_I_3 = 1.0430
REF_I = [REF_I_1, REF_I_3]

# parameter setting
N_RUNS_A = 100
A_BEAMLINE_LEN = 10

METHOD_OPTIONS = {
    "L-BFGS-B": {'eps': 1e-4, 'maxiter': 200, 'ftol': 1e-9, 'gtol': 1e-6},
    "SLSQP": {'eps': 1e-4, 'maxiter': 200, 'ftol': 1e-9},
    "trust-constr": {'finite_diff_rel_step': 1e-4, 'gtol': 1e-6, 'xtol': 1e-8,
    ↪ 'maxiter': 100},
    "Nelder-Mead": {'maxiter': 400, 'xatol': 1e-6, 'fatol': 1e-12},
    "COBYLA": {'maxiter': 400},
}

METHODS_A = [
    ("Nelder-Mead", None),
    ("L-BFGS-B", "2-point"),
    ("SLSQP", "2-point"),
    ("COBYLA", None),
    ("trust-constr", None)
]

A_VARS = {
    1: ["I", "current", lambda x: x],
    3: ["I2", "current", lambda x: x],
}

A_OBJ = {
    8: [{"measure": ["x", "alpha"], "goal": 0.0, "weight": 1.0}],
    9: [{"measure": ["y", "alpha"], "goal": 0.0, "weight": 1.0}],
}

```

time: 0 ns (started: 2026-06-02 16:39:08 -10:00)

```

[56]: %load_ext autotime

results_A, df_A, final_summary = run_scenario_A(CURRENT_BOUNDS, EPSILON,
    ↪ N_RUNS_A, REF_I, A_OBJ, A_BEAMLINE_LEN,
    ↪ METHODS_A, scale="log", METHOD_OPTIONS=METHOD_OPTIONS)
display_results(results_A)

```

The autotime extension is already loaded. To reload it, use:

```
%reload_ext autotime
```

Scenario A - Nelder-Mead (100 runs)

Nelder-Mead run 1/100 MSE=3.508e-06 nit=61 nfev=130 t=20.96s

Nelder-Mead	run 2/100	MSE=3.508e-06	nit=59	nfev=123	t=20.72s
Nelder-Mead	run 3/100	MSE=3.508e-06	nit=71	nfev=147	t=23.79s
Nelder-Mead	run 4/100	MSE=7.690e-13	nit=43	nfev=83	t=12.26s
Nelder-Mead	run 5/100	MSE=7.368e-13	nit=55	nfev=104	t=15.31s
Nelder-Mead	run 6/100	MSE=8.902e-13	nit=49	nfev=94	t=13.84s
Nelder-Mead	run 7/100	MSE=8.632e-13	nit=63	nfev=126	t=18.54s
Nelder-Mead	run 8/100	MSE=7.378e-13	nit=61	nfev=118	t=17.18s
Nelder-Mead	run 9/100	MSE=8.755e-14	nit=67	nfev=129	t=19.17s
Nelder-Mead	run 10/100	MSE=3.170e-14	nit=70	nfev=125	t=18.96s
Nelder-Mead	run 11/100	MSE=2.788e-14	nit=58	nfev=116	t=17.24s
Nelder-Mead	run 12/100	MSE=5.173e-13	nit=62	nfev=120	t=17.73s
Nelder-Mead	run 13/100	MSE=5.736e-13	nit=68	nfev=128	t=18.59s
Nelder-Mead	run 14/100	MSE=3.508e-06	nit=57	nfev=118	t=17.22s
Nelder-Mead	run 15/100	MSE=3.507e-06	nit=58	nfev=126	t=18.57s
Nelder-Mead	run 16/100	MSE=3.508e-06	nit=55	nfev=114	t=17.81s
Nelder-Mead	run 17/100	MSE=2.096e-13	nit=63	nfev=118	t=22.84s
Nelder-Mead	run 18/100	MSE=5.311e-13	nit=57	nfev=111	t=19.97s
Nelder-Mead	run 19/100	MSE=3.508e-06	nit=62	nfev=131	t=23.32s
Nelder-Mead	run 20/100	MSE=7.378e-13	nit=67	nfev=129	t=22.61s
Nelder-Mead	run 21/100	MSE=2.819e-13	nit=64	nfev=126	t=21.56s
Nelder-Mead	run 22/100	MSE=2.819e-13	nit=50	nfev=97	t=16.57s
Nelder-Mead	run 23/100	MSE=4.486e-13	nit=56	nfev=109	t=20.09s
Nelder-Mead	run 24/100	MSE=6.187e-13	nit=62	nfev=119	t=24.97s
Nelder-Mead	run 25/100	MSE=2.088e-13	nit=69	nfev=136	t=27.13s
Nelder-Mead	run 26/100	MSE=1.541e+01	nit=39	nfev=85	t=15.18s
Nelder-Mead	run 27/100	MSE=2.743e-13	nit=67	nfev=122	t=23.68s
Nelder-Mead	run 28/100	MSE=1.437e-12	nit=54	nfev=102	t=19.65s
Nelder-Mead	run 29/100	MSE=1.694e-13	nit=56	nfev=105	t=20.41s
Nelder-Mead	run 30/100	MSE=2.819e-13	nit=50	nfev=96	t=18.75s
Nelder-Mead	run 31/100	MSE=3.508e-06	nit=54	nfev=110	t=21.04s
Nelder-Mead	run 32/100	MSE=6.498e-13	nit=57	nfev=111	t=23.61s
Nelder-Mead	run 33/100	MSE=3.507e-06	nit=50	nfev=110	t=20.58s
Nelder-Mead	run 34/100	MSE=7.221e-13	nit=58	nfev=110	t=21.42s
Nelder-Mead	run 35/100	MSE=1.067e-13	nit=61	nfev=119	t=23.52s
Nelder-Mead	run 36/100	MSE=3.508e-06	nit=56	nfev=115	t=24.26s
Nelder-Mead	run 37/100	MSE=5.573e-13	nit=61	nfev=118	t=25.21s
Nelder-Mead	run 38/100	MSE=1.482e-13	nit=76	nfev=144	t=29.50s
Nelder-Mead	run 39/100	MSE=6.654e-13	nit=58	nfev=112	t=23.45s
Nelder-Mead	run 40/100	MSE=3.508e-06	nit=71	nfev=142	t=28.44s
Nelder-Mead	run 41/100	MSE=1.448e-12	nit=50	nfev=99	t=19.92s
Nelder-Mead	run 42/100	MSE=1.730e-12	nit=65	nfev=123	t=25.90s
Nelder-Mead	run 43/100	MSE=7.365e-13	nit=57	nfev=110	t=22.29s
Nelder-Mead	run 44/100	MSE=9.370e-02	nit=13	nfev=25	t=5.30s
Nelder-Mead	run 45/100	MSE=3.508e-06	nit=55	nfev=110	t=23.12s
Nelder-Mead	run 46/100	MSE=3.508e-06	nit=73	nfev=153	t=31.68s
Nelder-Mead	run 47/100	MSE=1.110e-12	nit=52	nfev=104	t=20.83s
Nelder-Mead	run 48/100	MSE=3.507e-06	nit=64	nfev=133	t=27.51s
Nelder-Mead	run 49/100	MSE=4.107e-13	nit=72	nfev=137	t=28.15s

Nelder-Mead	run 50/100	MSE=3.170e-14	nit=66	nfev=124	t=25.56s
Nelder-Mead	run 51/100	MSE=3.508e-06	nit=53	nfev=110	t=28.88s
Nelder-Mead	run 52/100	MSE=4.201e-13	nit=48	nfev=91	t=18.40s
Nelder-Mead	run 53/100	MSE=3.418e-13	nit=62	nfev=116	t=24.87s
Nelder-Mead	run 54/100	MSE=2.655e-13	nit=65	nfev=121	t=28.68s
Nelder-Mead	run 55/100	MSE=1.057e-13	nit=66	nfev=124	t=23.69s
Nelder-Mead	run 56/100	MSE=4.486e-13	nit=45	nfev=87	t=19.23s
Nelder-Mead	run 57/100	MSE=3.508e-06	nit=56	nfev=123	t=26.74s
Nelder-Mead	run 58/100	MSE=2.819e-13	nit=55	nfev=105	t=22.59s
Nelder-Mead	run 59/100	MSE=2.847e-14	nit=65	nfev=122	t=25.35s
Nelder-Mead	run 60/100	MSE=2.165e-13	nit=65	nfev=128	t=26.88s
Nelder-Mead	run 61/100	MSE=4.554e-13	nit=61	nfev=118	t=24.93s
Nelder-Mead	run 62/100	MSE=2.819e-13	nit=51	nfev=98	t=21.00s
Nelder-Mead	run 63/100	MSE=3.508e-06	nit=55	nfev=119	t=24.42s
Nelder-Mead	run 64/100	MSE=5.246e-13	nit=67	nfev=125	t=28.64s
Nelder-Mead	run 65/100	MSE=2.888e-13	nit=64	nfev=122	t=24.79s
Nelder-Mead	run 66/100	MSE=3.508e-06	nit=64	nfev=134	t=26.83s
Nelder-Mead	run 67/100	MSE=4.571e-13	nit=74	nfev=144	t=33.61s
Nelder-Mead	run 68/100	MSE=9.896e-13	nit=50	nfev=97	t=32.65s
Nelder-Mead	run 69/100	MSE=3.508e-06	nit=74	nfev=146	t=33.26s
Nelder-Mead	run 70/100	MSE=3.508e-06	nit=58	nfev=116	t=24.86s
Nelder-Mead	run 71/100	MSE=9.370e-02	nit=16	nfev=31	t=6.95s
Nelder-Mead	run 72/100	MSE=9.370e-02	nit=13	nfev=25	t=5.00s
Nelder-Mead	run 73/100	MSE=1.080e-13	nit=57	nfev=112	t=21.52s
Nelder-Mead	run 74/100	MSE=6.074e-13	nit=52	nfev=101	t=19.30s
Nelder-Mead	run 75/100	MSE=3.508e-06	nit=57	nfev=119	t=23.83s
Nelder-Mead	run 76/100	MSE=2.819e-13	nit=59	nfev=112	t=24.37s
Nelder-Mead	run 77/100	MSE=2.819e-13	nit=55	nfev=101	t=21.07s
Nelder-Mead	run 78/100	MSE=4.993e-13	nit=68	nfev=130	t=24.82s
Nelder-Mead	run 79/100	MSE=8.350e-13	nit=50	nfev=96	t=18.67s
Nelder-Mead	run 80/100	MSE=9.367e-02	nit=36	nfev=72	t=14.01s
Nelder-Mead	run 81/100	MSE=3.507e-06	nit=63	nfev=135	t=29.74s
Nelder-Mead	run 82/100	MSE=1.229e-12	nit=47	nfev=91	t=19.03s
Nelder-Mead	run 83/100	MSE=3.508e-06	nit=64	nfev=138	t=29.78s
Nelder-Mead	run 84/100	MSE=3.508e-06	nit=58	nfev=117	t=34.44s
Nelder-Mead	run 85/100	MSE=6.187e-13	nit=64	nfev=122	t=24.14s
Nelder-Mead	run 86/100	MSE=8.785e-14	nit=64	nfev=121	t=24.36s
Nelder-Mead	run 87/100	MSE=3.507e-06	nit=63	nfev=133	t=33.92s
Nelder-Mead	run 88/100	MSE=8.035e-13	nit=49	nfev=94	t=21.50s
Nelder-Mead	run 89/100	MSE=1.142e-13	nit=62	nfev=120	t=24.26s
Nelder-Mead	run 90/100	MSE=3.508e-06	nit=63	nfev=132	t=28.11s
Nelder-Mead	run 91/100	MSE=4.457e-13	nit=56	nfev=106	t=21.86s
Nelder-Mead	run 92/100	MSE=1.610e-12	nit=56	nfev=110	t=22.49s
Nelder-Mead	run 93/100	MSE=3.508e-06	nit=62	nfev=132	t=29.88s
Nelder-Mead	run 94/100	MSE=4.107e-13	nit=57	nfev=110	t=23.30s
Nelder-Mead	run 95/100	MSE=3.508e-06	nit=55	nfev=121	t=25.06s
Nelder-Mead	run 96/100	MSE=1.256e-12	nit=59	nfev=113	t=22.35s
Nelder-Mead	run 97/100	MSE=3.508e-06	nit=61	nfev=129	t=24.63s

Nelder-Mead	run 98/100	MSE=4.486e-13	nit=53	nfev=100	t=19.64s
Nelder-Mead	run 99/100	MSE=3.508e-06	nit=58	nfev=119	t=23.75s
Nelder-Mead	run 100/100	MSE=3.170e-14	nit=51	nfev=95	t=18.36s

Scenario A - L-BFGS-B+jac (100 runs)

L-BFGS-B	run 1/100	MSE=1.247e-06	nit=12	nfev=231	t=58.53s
L-BFGS-B	run 2/100	MSE=6.835e-05	nit=15	nfev=99	t=21.07s
L-BFGS-B	run 3/100	MSE=5.554e-04	nit=22	nfev=144	t=34.35s
L-BFGS-B	run 4/100	MSE=1.374e-05	nit=8	nfev=138	t=33.12s
L-BFGS-B	run 5/100	MSE=3.334e-05	nit=16	nfev=186	t=39.40s
L-BFGS-B	run 6/100	MSE=6.811e-05	nit=8	nfev=87	t=18.09s
L-BFGS-B	run 7/100	MSE=1.025e-04	nit=13	nfev=171	t=34.77s
L-BFGS-B	run 8/100	MSE=9.367e-02	nit=4	nfev=75	t=16.66s
L-BFGS-B	run 9/100	MSE=2.513e-04	nit=11	nfev=66	t=13.95s
L-BFGS-B	run 10/100	MSE=4.378e-05	nit=15	nfev=153	t=32.07s
L-BFGS-B	run 11/100	MSE=2.510e-04	nit=9	nfev=72	t=14.06s
L-BFGS-B	run 12/100	MSE=2.362e-04	nit=11	nfev=96	t=18.22s
L-BFGS-B	run 13/100	MSE=1.589e-05	nit=15	nfev=240	t=46.36s
L-BFGS-B	run 14/100	MSE=8.824e-06	nit=14	nfev=159	t=30.63s
L-BFGS-B	run 15/100	MSE=9.367e-02	nit=4	nfev=102	t=19.74s
L-BFGS-B	run 16/100	MSE=5.555e-04	nit=10	nfev=72	t=13.83s
L-BFGS-B	run 17/100	MSE=7.333e-05	nit=11	nfev=153	t=29.61s
L-BFGS-B	run 18/100	MSE=1.764e-05	nit=15	nfev=183	t=35.26s
L-BFGS-B	run 19/100	MSE=1.343e-05	nit=18	nfev=384	t=75.18s
L-BFGS-B	run 20/100	MSE=9.367e-02	nit=5	nfev=99	t=20.13s
L-BFGS-B	run 21/100	MSE=3.515e-05	nit=11	nfev=147	t=30.78s
L-BFGS-B	run 22/100	MSE=2.583e-05	nit=12	nfev=264	t=51.80s
L-BFGS-B	run 23/100	MSE=1.836e-06	nit=14	nfev=222	t=52.01s
L-BFGS-B	run 24/100	MSE=3.326e-05	nit=14	nfev=297	t=114.72s
L-BFGS-B	run 25/100	MSE=2.499e-04	nit=15	nfev=126	t=26.75s
L-BFGS-B	run 26/100	MSE=9.556e-05	nit=7	nfev=231	t=62.10s
L-BFGS-B	run 27/100	MSE=2.393e-06	nit=17	nfev=195	t=40.74s
L-BFGS-B	run 28/100	MSE=3.235e-05	nit=13	nfev=204	t=43.90s
L-BFGS-B	run 29/100	MSE=3.652e-05	nit=16	nfev=213	t=44.01s
L-BFGS-B	run 30/100	MSE=1.747e-06	nit=10	nfev=237	t=52.09s
L-BFGS-B	run 31/100	MSE=5.550e-04	nit=14	nfev=153	t=32.54s
L-BFGS-B	run 32/100	MSE=2.419e-05	nit=11	nfev=63	t=13.55s
L-BFGS-B	run 33/100	MSE=5.556e-04	nit=10	nfev=78	t=16.44s
L-BFGS-B	run 34/100	MSE=2.322e-06	nit=14	nfev=261	t=58.01s
L-BFGS-B	run 35/100	MSE=2.321e-04	nit=17	nfev=123	t=25.29s
L-BFGS-B	run 36/100	MSE=4.910e-06	nit=14	nfev=201	t=42.09s
L-BFGS-B	run 37/100	MSE=2.372e-04	nit=12	nfev=93	t=17.12s
L-BFGS-B	run 38/100	MSE=4.344e-05	nit=15	nfev=258	t=46.82s
L-BFGS-B	run 39/100	MSE=1.673e-04	nit=13	nfev=183	t=34.09s
L-BFGS-B	run 40/100	MSE=5.613e-06	nit=18	nfev=411	t=75.68s
L-BFGS-B	run 41/100	MSE=9.367e-02	nit=4	nfev=96	t=18.72s
L-BFGS-B	run 42/100	MSE=6.507e-05	nit=11	nfev=108	t=20.61s
L-BFGS-B	run 43/100	MSE=4.844e-05	nit=6	nfev=66	t=11.85s

L-BFGS-B	run 44/100	MSE=7.514e-05	nit=14	nfev=231	t=42.66s
L-BFGS-B	run 45/100	MSE=5.546e-04	nit=16	nfev=138	t=25.43s
L-BFGS-B	run 46/100	MSE=3.268e-05	nit=11	nfev=156	t=28.49s
L-BFGS-B	run 47/100	MSE=9.367e-02	nit=4	nfev=126	t=22.63s
L-BFGS-B	run 48/100	MSE=5.683e-06	nit=17	nfev=219	t=40.15s
L-BFGS-B	run 49/100	MSE=6.677e-05	nit=10	nfev=177	t=32.85s
L-BFGS-B	run 50/100	MSE=4.740e-05	nit=15	nfev=159	t=29.06s
L-BFGS-B	run 51/100	MSE=5.559e-04	nit=18	nfev=153	t=28.46s
L-BFGS-B	run 52/100	MSE=1.953e-05	nit=16	nfev=162	t=34.11s
L-BFGS-B	run 53/100	MSE=2.247e-05	nit=14	nfev=213	t=46.07s
L-BFGS-B	run 54/100	MSE=5.560e-04	nit=20	nfev=111	t=24.70s
L-BFGS-B	run 55/100	MSE=6.628e-05	nit=14	nfev=96	t=23.69s
L-BFGS-B	run 56/100	MSE=9.367e-02	nit=4	nfev=66	t=14.47s
L-BFGS-B	run 57/100	MSE=9.367e-02	nit=4	nfev=78	t=17.16s
L-BFGS-B	run 58/100	MSE=9.576e-06	nit=11	nfev=156	t=31.79s
L-BFGS-B	run 59/100	MSE=8.266e-06	nit=21	nfev=264	t=53.43s
L-BFGS-B	run 60/100	MSE=2.349e-04	nit=19	nfev=147	t=28.26s
L-BFGS-B	run 61/100	MSE=2.474e-04	nit=11	nfev=90	t=18.12s
L-BFGS-B	run 62/100	MSE=9.367e-02	nit=4	nfev=66	t=13.74s
L-BFGS-B	run 63/100	MSE=1.844e-05	nit=16	nfev=117	t=20.11s
L-BFGS-B	run 64/100	MSE=5.477e-06	nit=17	nfev=216	t=35.95s
L-BFGS-B	run 65/100	MSE=1.457e-04	nit=20	nfev=219	t=35.18s
L-BFGS-B	run 66/100	MSE=3.588e-06	nit=13	nfev=165	t=25.70s
L-BFGS-B	run 67/100	MSE=1.382e-05	nit=14	nfev=255	t=39.98s
L-BFGS-B	run 68/100	MSE=6.492e-05	nit=9	nfev=99	t=15.68s
L-BFGS-B	run 69/100	MSE=3.332e-06	nit=10	nfev=159	t=24.74s
L-BFGS-B	run 70/100	MSE=5.557e-04	nit=14	nfev=96	t=15.15s
L-BFGS-B	run 71/100	MSE=2.644e-05	nit=17	nfev=312	t=50.16s
L-BFGS-B	run 72/100	MSE=4.250e-05	nit=19	nfev=204	t=36.68s
L-BFGS-B	run 73/100	MSE=2.491e-04	nit=14	nfev=135	t=34.31s
L-BFGS-B	run 74/100	MSE=6.041e-05	nit=15	nfev=153	t=26.08s
L-BFGS-B	run 75/100	MSE=2.603e-04	nit=7	nfev=159	t=28.19s
L-BFGS-B	run 76/100	MSE=3.078e-05	nit=12	nfev=87	t=14.37s
L-BFGS-B	run 77/100	MSE=5.559e-04	nit=16	nfev=117	t=19.45s
L-BFGS-B	run 78/100	MSE=2.663e-05	nit=10	nfev=231	t=38.02s
L-BFGS-B	run 79/100	MSE=5.945e-05	nit=14	nfev=177	t=29.35s
L-BFGS-B	run 80/100	MSE=2.507e-04	nit=15	nfev=96	t=16.07s
L-BFGS-B	run 81/100	MSE=3.566e-06	nit=14	nfev=108	t=17.75s
L-BFGS-B	run 82/100	MSE=1.213e-05	nit=9	nfev=261	t=44.66s
L-BFGS-B	run 83/100	MSE=3.331e-05	nit=8	nfev=228	t=40.26s
L-BFGS-B	run 84/100	MSE=5.554e-04	nit=13	nfev=78	t=15.00s
L-BFGS-B	run 85/100	MSE=6.846e-05	nit=10	nfev=102	t=18.18s
L-BFGS-B	run 86/100	MSE=5.613e-05	nit=15	nfev=333	t=59.97s
L-BFGS-B	run 87/100	MSE=1.968e-04	nit=10	nfev=126	t=22.69s
L-BFGS-B	run 88/100	MSE=2.506e-04	nit=11	nfev=93	t=16.92s
L-BFGS-B	run 89/100	MSE=5.504e-04	nit=24	nfev=147	t=24.65s
L-BFGS-B	run 90/100	MSE=5.149e-06	nit=13	nfev=369	t=62.34s
L-BFGS-B	run 91/100	MSE=2.459e-04	nit=13	nfev=93	t=15.73s

L-BFGS-B	run 92/100	MSE=6.597e-05	nit=8	nfev=138	t=24.06s
L-BFGS-B	run 93/100	MSE=4.618e-05	nit=18	nfev=183	t=31.74s
L-BFGS-B	run 94/100	MSE=9.720e-06	nit=14	nfev=249	t=41.53s
L-BFGS-B	run 95/100	MSE=1.721e-05	nit=14	nfev=315	t=52.25s
L-BFGS-B	run 96/100	MSE=1.734e-05	nit=11	nfev=156	t=25.71s
L-BFGS-B	run 97/100	MSE=1.614e-05	nit=15	nfev=252	t=39.70s
L-BFGS-B	run 98/100	MSE=9.367e-02	nit=9	nfev=204	t=32.96s
L-BFGS-B	run 99/100	MSE=6.418e-06	nit=14	nfev=204	t=32.92s
L-BFGS-B	run 100/100	MSE=1.842e-05	nit=8	nfev=87	t=13.84s

Scenario A - SLSQP+jac (100 runs)

SLSQP	run 1/100	MSE=6.791e-05	nit=14	nfev=58	t=9.33s
SLSQP	run 2/100	MSE=6.786e-06	nit=36	nfev=380	t=61.17s
SLSQP	run 3/100	MSE=4.484e-06	nit=44	nfev=458	t=74.03s
SLSQP	run 4/100	MSE=5.131e-05	nit=18	nfev=179	t=29.27s
SLSQP	run 5/100	MSE=1.588e-04	nit=18	nfev=76	t=12.40s
SLSQP	run 6/100	MSE=5.310e-05	nit=7	nfev=36	t=5.76s
SLSQP	run 7/100	MSE=5.547e-04	nit=13	nfev=52	t=8.47s
SLSQP	run 8/100	MSE=9.367e-02	nit=6	nfev=18	t=3.15s
SLSQP	run 9/100	MSE=1.052e-04	nit=13	nfev=108	t=17.32s
SLSQP	run 10/100	MSE=1.285e-04	nit=17	nfev=79	t=13.11s
SLSQP	run 11/100	MSE=2.489e-04	nit=6	nfev=31	t=5.20s
SLSQP	run 12/100	MSE=2.402e-07	nit=78	nfev=957	t=158.53s
SLSQP	run 13/100	MSE=3.199e-05	nit=13	nfev=53	t=8.64s
SLSQP	run 14/100	MSE=1.181e-05	nit=200	nfev=2505	t=417.91s
SLSQP	run 15/100	MSE=9.367e-02	nit=7	nfev=40	t=6.53s
SLSQP	run 16/100	MSE=8.564e-05	nit=46	nfev=574	t=95.64s
SLSQP	run 17/100	MSE=3.280e-05	nit=12	nfev=84	t=13.26s
SLSQP	run 18/100	MSE=5.559e-04	nit=17	nfev=55	t=9.00s
SLSQP	run 19/100	MSE=1.751e-05	nit=13	nfev=65	t=10.40s
SLSQP	run 20/100	MSE=9.367e-02	nit=10	nfev=48	t=7.67s
SLSQP	run 21/100	MSE=5.528e-04	nit=13	nfev=53	t=8.63s
SLSQP	run 22/100	MSE=6.750e-05	nit=10	nfev=45	t=7.31s
SLSQP	run 23/100	MSE=3.825e-05	nit=9	nfev=41	t=6.76s
SLSQP	run 24/100	MSE=1.629e-05	nit=12	nfev=69	t=11.35s
SLSQP	run 25/100	MSE=5.564e-04	nit=20	nfev=67	t=11.07s
SLSQP	run 26/100	MSE=2.511e-04	nit=18	nfev=61	t=9.91s
SLSQP	run 27/100	MSE=2.511e-04	nit=12	nfev=38	t=6.04s
SLSQP	run 28/100	MSE=9.367e-02	nit=8	nfev=41	t=7.06s
SLSQP	run 29/100	MSE=1.660e-04	nit=12	nfev=68	t=11.84s
SLSQP	run 30/100	MSE=7.204e-06	nit=12	nfev=71	t=11.97s
SLSQP	run 31/100	MSE=5.559e-04	nit=13	nfev=43	t=6.83s
SLSQP	run 32/100	MSE=3.669e-05	nit=11	nfev=54	t=8.87s
SLSQP	run 33/100	MSE=2.180e-04	nit=6	nfev=42	t=6.98s
SLSQP	run 34/100	MSE=1.893e-04	nit=16	nfev=64	t=10.56s
SLSQP	run 35/100	MSE=9.692e-06	nit=200	nfev=2487	t=416.87s
SLSQP	run 36/100	MSE=1.912e-05	nit=133	nfev=1644	t=285.98s
SLSQP	run 37/100	MSE=9.367e-02	nit=7	nfev=37	t=6.23s

SLSQP	run 38/100	MSE=5.562e-04	nit=16	nfev=51	t=8.57s
SLSQP	run 39/100	MSE=2.492e-05	nit=85	nfev=1041	t=176.58s
SLSQP	run 40/100	MSE=5.386e-06	nit=200	nfev=2525	t=434.67s
SLSQP	run 41/100	MSE=4.345e-06	nit=7	nfev=46	t=7.66s
SLSQP	run 42/100	MSE=5.560e-04	nit=13	nfev=42	t=7.12s
SLSQP	run 43/100	MSE=2.497e-04	nit=7	nfev=34	t=5.69s
SLSQP	run 44/100	MSE=5.563e-04	nit=18	nfev=61	t=10.17s
SLSQP	run 45/100	MSE=5.562e-04	nit=15	nfev=50	t=8.73s
SLSQP	run 46/100	MSE=2.837e-05	nit=9	nfev=41	t=6.60s
SLSQP	run 47/100	MSE=9.367e-02	nit=6	nfev=29	t=4.88s
SLSQP	run 48/100	MSE=1.967e-04	nit=13	nfev=53	t=9.15s
SLSQP	run 49/100	MSE=2.320e-04	nit=14	nfev=59	t=10.00s
SLSQP	run 50/100	MSE=3.934e-06	nit=200	nfev=2432	t=423.23s
SLSQP	run 51/100	MSE=4.710e-04	nit=8	nfev=37	t=6.14s
SLSQP	run 52/100	MSE=5.140e-05	nit=15	nfev=92	t=15.79s
SLSQP	run 53/100	MSE=1.788e-08	nit=93	nfev=1047	t=174.18s
SLSQP	run 54/100	MSE=5.548e-04	nit=13	nfev=52	t=8.03s
SLSQP	run 55/100	MSE=1.000e-05	nit=200	nfev=2431	t=380.66s
SLSQP	run 56/100	MSE=9.367e-02	nit=5	nfev=35	t=5.76s
SLSQP	run 57/100	MSE=6.363e-05	nit=16	nfev=98	t=16.38s
SLSQP	run 58/100	MSE=4.516e-05	nit=13	nfev=80	t=13.40s
SLSQP	run 59/100	MSE=6.471e-05	nit=16	nfev=62	t=10.42s
SLSQP	run 60/100	MSE=9.367e-02	nit=10	nfev=48	t=8.24s
SLSQP	run 61/100	MSE=2.511e-04	nit=11	nfev=37	t=6.46s
SLSQP	run 62/100	MSE=9.367e-02	nit=5	nfev=19	t=3.02s
SLSQP	run 63/100	MSE=1.607e-08	nit=84	nfev=961	t=163.30s
SLSQP	run 64/100	MSE=5.092e-06	nit=12	nfev=61	t=9.85s
SLSQP	run 65/100	MSE=9.367e-02	nit=10	nfev=48	t=7.68s
SLSQP	run 66/100	MSE=4.688e-05	nit=10	nfev=45	t=7.13s
SLSQP	run 67/100	MSE=1.463e-04	nit=12	nfev=58	t=8.97s
SLSQP	run 68/100	MSE=6.767e-05	nit=7	nfev=35	t=5.73s
SLSQP	run 69/100	MSE=7.703e-05	nit=10	nfev=62	t=10.11s
SLSQP	run 70/100	MSE=1.102e-05	nit=200	nfev=2528	t=423.87s
SLSQP	run 71/100	MSE=6.659e-05	nit=20	nfev=77	t=13.37s
SLSQP	run 72/100	MSE=2.507e-04	nit=20	nfev=80	t=15.38s
SLSQP	run 73/100	MSE=1.313e-05	nit=20	nfev=214	t=37.03s
SLSQP	run 74/100	MSE=9.337e-05	nit=21	nfev=111	t=19.02s
SLSQP	run 75/100	MSE=5.555e-04	nit=12	nfev=51	t=8.62s
SLSQP	run 76/100	MSE=2.349e-06	nit=49	nfev=573	t=99.58s
SLSQP	run 77/100	MSE=3.736e-06	nit=25	nfev=269	t=45.64s
SLSQP	run 78/100	MSE=2.166e-04	nit=19	nfev=70	t=12.71s
SLSQP	run 79/100	MSE=1.369e-05	nit=20	nfev=175	t=29.06s
SLSQP	run 80/100	MSE=9.367e-02	nit=16	nfev=83	t=14.35s
SLSQP	run 81/100	MSE=5.460e-05	nit=14	nfev=60	t=9.94s
SLSQP	run 82/100	MSE=6.858e-05	nit=9	nfev=31	t=7.22s
SLSQP	run 83/100	MSE=4.846e-06	nit=200	nfev=2501	t=558.48s
SLSQP	run 84/100	MSE=5.561e-04	nit=9	nfev=29	t=4.67s
SLSQP	run 85/100	MSE=5.560e-04	nit=14	nfev=46	t=8.26s

SLSQP	run 86/100	MSE=6.788e-05	nit=10	nfev=47	t=7.79s
SLSQP	run 87/100	MSE=1.257e-04	nit=8	nfev=41	t=6.64s
SLSQP	run 88/100	MSE=1.281e-04	nit=7	nfev=35	t=5.46s
SLSQP	run 89/100	MSE=9.433e-06	nit=200	nfev=2469	t=457.03s
SLSQP	run 90/100	MSE=5.555e-04	nit=12	nfev=40	t=6.27s
SLSQP	run 91/100	MSE=6.373e-05	nit=6	nfev=31	t=4.85s
SLSQP	run 92/100	MSE=4.686e-06	nit=10	nfev=56	t=8.69s
SLSQP	run 93/100	MSE=8.234e-06	nit=200	nfev=2472	t=389.30s
SLSQP	run 94/100	MSE=9.367e-02	nit=6	nfev=30	t=4.86s
SLSQP	run 95/100	MSE=5.553e-04	nit=11	nfev=35	t=6.65s
SLSQP	run 96/100	MSE=9.957e-06	nit=12	nfev=71	t=11.17s
SLSQP	run 97/100	MSE=4.708e-05	nit=15	nfev=73	t=12.26s
SLSQP	run 98/100	MSE=9.787e-07	nit=65	nfev=782	t=154.54s
SLSQP	run 99/100	MSE=1.038e-06	nit=14	nfev=70	t=12.67s
SLSQP	run 100/100	MSE=6.880e-05	nit=8	nfev=29	t=5.54s

Scenario A - COBYLA (100 runs)

COBYLA	run 1/100	MSE=4.009e-05	nit=-1	nfev=182	t=35.30s
COBYLA	run 2/100	MSE=9.346e-02	nit=-1	nfev=45	t=8.46s
COBYLA	run 3/100	MSE=4.746e-04	nit=-1	nfev=132	t=25.18s
COBYLA	run 4/100	MSE=4.095e-03	nit=-1	nfev=400	t=69.76s
COBYLA	run 5/100	MSE=2.635e-03	nit=-1	nfev=400	t=76.03s
COBYLA	run 6/100	MSE=2.141e-03	nit=-1	nfev=400	t=81.47s
COBYLA	run 7/100	MSE=2.103e-04	nit=-1	nfev=189	t=42.74s
COBYLA	run 8/100	MSE=6.885e-04	nit=-1	nfev=119	t=25.18s
COBYLA	run 9/100	MSE=5.018e-03	nit=-1	nfev=400	t=147.89s
COBYLA	run 10/100	MSE=2.281e-04	nit=-1	nfev=177	t=69.76s
COBYLA	run 11/100	MSE=9.320e-02	nit=-1	nfev=77	t=28.06s
COBYLA	run 12/100	MSE=1.859e-03	nit=-1	nfev=400	t=139.36s
COBYLA	run 13/100	MSE=2.653e-04	nit=-1	nfev=167	t=81.81s
COBYLA	run 14/100	MSE=5.099e-06	nit=-1	nfev=83	t=29.53s
COBYLA	run 15/100	MSE=4.880e-05	nit=-1	nfev=31	t=12.69s
COBYLA	run 16/100	MSE=3.002e-04	nit=-1	nfev=116	t=57.97s
COBYLA	run 17/100	MSE=1.224e-03	nit=-1	nfev=400	t=185.29s
COBYLA	run 18/100	MSE=2.872e-04	nit=-1	nfev=134	t=48.15s
COBYLA	run 19/100	MSE=9.346e-02	nit=-1	nfev=45	t=14.60s
COBYLA	run 20/100	MSE=5.625e-05	nit=-1	nfev=32	t=8.43s
COBYLA	run 21/100	MSE=3.144e-05	nit=-1	nfev=287	t=108.37s
COBYLA	run 22/100	MSE=1.700e-04	nit=-1	nfev=166	t=43.06s
COBYLA	run 23/100	MSE=2.108e-04	nit=-1	nfev=193	t=67.23s
COBYLA	run 24/100	MSE=3.645e-04	nit=-1	nfev=127	t=40.68s
COBYLA	run 25/100	MSE=4.411e-03	nit=-1	nfev=400	t=106.61s
COBYLA	run 26/100	MSE=3.807e-05	nit=-1	nfev=84	t=33.86s
COBYLA	run 27/100	MSE=1.016e-04	nit=-1	nfev=208	t=64.51s
COBYLA	run 28/100	MSE=2.761e-04	nit=-1	nfev=164	t=66.95s
COBYLA	run 29/100	MSE=1.859e-03	nit=-1	nfev=400	t=148.22s
COBYLA	run 30/100	MSE=2.937e-04	nit=-1	nfev=146	t=52.00s
COBYLA	run 31/100	MSE=3.844e-04	nit=-1	nfev=146	t=41.45s

COBYLA	run 32/100	MSE=3.133e-04	nit=-1	nfev=205	t=67.49s
COBYLA	run 33/100	MSE=1.299e-04	nit=-1	nfev=148	t=62.43s
COBYLA	run 34/100	MSE=2.700e-04	nit=-1	nfev=242	t=71.90s
COBYLA	run 35/100	MSE=3.773e-04	nit=-1	nfev=169	t=57.69s
COBYLA	run 36/100	MSE=8.143e-06	nit=-1	nfev=82	t=29.87s
COBYLA	run 37/100	MSE=1.271e-03	nit=-1	nfev=400	t=95.71s
COBYLA	run 38/100	MSE=1.002e-04	nit=-1	nfev=30	t=9.86s
COBYLA	run 39/100	MSE=9.346e-02	nit=-1	nfev=39	t=11.01s
COBYLA	run 40/100	MSE=1.529e-06	nit=-1	nfev=84	t=32.03s
COBYLA	run 41/100	MSE=2.802e-06	nit=-1	nfev=88	t=31.28s
COBYLA	run 42/100	MSE=9.346e-02	nit=-1	nfev=44	t=13.07s
COBYLA	run 43/100	MSE=9.320e-02	nit=-1	nfev=77	t=32.93s
COBYLA	run 44/100	MSE=2.325e-03	nit=-1	nfev=400	t=124.86s
COBYLA	run 45/100	MSE=2.625e-04	nit=-1	nfev=236	t=80.18s
COBYLA	run 46/100	MSE=1.606e-04	nit=-1	nfev=269	t=101.78s
COBYLA	run 47/100	MSE=3.135e-04	nit=-1	nfev=197	t=51.84s
COBYLA	run 48/100	MSE=9.901e-02	nit=-1	nfev=30	t=11.13s
COBYLA	run 49/100	MSE=9.346e-02	nit=-1	nfev=44	t=14.32s
COBYLA	run 50/100	MSE=3.653e-04	nit=-1	nfev=191	t=76.86s
COBYLA	run 51/100	MSE=2.239e-05	nit=-1	nfev=217	t=86.00s
COBYLA	run 52/100	MSE=2.272e-04	nit=-1	nfev=103	t=36.73s
COBYLA	run 53/100	MSE=1.935e-04	nit=-1	nfev=135	t=46.57s
COBYLA	run 54/100	MSE=6.144e-04	nit=-1	nfev=58	t=18.67s
COBYLA	run 55/100	MSE=1.709e-05	nit=-1	nfev=82	t=26.53s
COBYLA	run 56/100	MSE=3.514e-04	nit=-1	nfev=400	t=127.47s
COBYLA	run 57/100	MSE=5.622e-06	nit=-1	nfev=65	t=21.62s
COBYLA	run 58/100	MSE=3.108e-03	nit=-1	nfev=400	t=112.66s
COBYLA	run 59/100	MSE=4.016e-04	nit=-1	nfev=120	t=30.67s
COBYLA	run 60/100	MSE=2.041e-04	nit=-1	nfev=180	t=49.42s
COBYLA	run 61/100	MSE=1.530e-03	nit=-1	nfev=400	t=117.46s
COBYLA	run 62/100	MSE=1.525e-04	nit=-1	nfev=103	t=30.20s
COBYLA	run 63/100	MSE=5.295e-06	nit=-1	nfev=122	t=33.91s
COBYLA	run 64/100	MSE=1.647e-04	nit=-1	nfev=218	t=62.83s
COBYLA	run 65/100	MSE=2.699e-04	nit=-1	nfev=125	t=35.86s
COBYLA	run 66/100	MSE=2.198e-06	nit=-1	nfev=129	t=38.51s
COBYLA	run 67/100	MSE=9.346e-02	nit=-1	nfev=44	t=14.61s
COBYLA	run 68/100	MSE=1.834e-03	nit=-1	nfev=400	t=117.35s
COBYLA	run 69/100	MSE=9.346e-02	nit=-1	nfev=45	t=13.70s
COBYLA	run 70/100	MSE=3.621e-06	nit=-1	nfev=29	t=8.24s
COBYLA	run 71/100	MSE=3.421e-04	nit=-1	nfev=124	t=49.09s
COBYLA	run 72/100	MSE=2.462e-04	nit=-1	nfev=173	t=47.67s
COBYLA	run 73/100	MSE=1.859e-03	nit=-1	nfev=400	t=119.70s
COBYLA	run 74/100	MSE=5.415e-04	nit=-1	nfev=131	t=40.82s
COBYLA	run 75/100	MSE=3.592e-04	nit=-1	nfev=198	t=61.66s
COBYLA	run 76/100	MSE=4.265e-06	nit=-1	nfev=106	t=54.68s
COBYLA	run 77/100	MSE=2.654e-05	nit=-1	nfev=75	t=22.48s
COBYLA	run 78/100	MSE=1.859e-03	nit=-1	nfev=400	t=130.69s
COBYLA	run 79/100	MSE=2.593e-04	nit=-1	nfev=155	t=50.85s

COBYLA	run 80/100	MSE=2.592e-04	nit=-1	nfev=102	t=28.27s
COBYLA	run 81/100	MSE=7.878e-06	nit=-1	nfev=118	t=40.78s
COBYLA	run 82/100	MSE=1.119e-03	nit=-1	nfev=400	t=165.55s
COBYLA	run 83/100	MSE=1.197e-04	nit=-1	nfev=239	t=74.40s
COBYLA	run 84/100	MSE=8.235e-05	nit=-1	nfev=193	t=43.34s
COBYLA	run 85/100	MSE=9.346e-02	nit=-1	nfev=40	t=10.99s
COBYLA	run 86/100	MSE=1.224e-03	nit=-1	nfev=400	t=149.36s
COBYLA	run 87/100	MSE=5.060e-06	nit=-1	nfev=26	t=8.96s
COBYLA	run 88/100	MSE=1.524e-03	nit=-1	nfev=400	t=142.84s
COBYLA	run 89/100	MSE=1.881e-04	nit=-1	nfev=209	t=73.28s
COBYLA	run 90/100	MSE=2.272e-04	nit=-1	nfev=200	t=74.09s
COBYLA	run 91/100	MSE=1.859e-03	nit=-1	nfev=400	t=101.91s
COBYLA	run 92/100	MSE=1.781e-02	nit=-1	nfev=400	t=91.25s
COBYLA	run 93/100	MSE=2.759e-05	nit=-1	nfev=52	t=11.63s
COBYLA	run 94/100	MSE=3.733e-04	nit=-1	nfev=153	t=31.39s
COBYLA	run 95/100	MSE=8.438e-06	nit=-1	nfev=115	t=26.73s
COBYLA	run 96/100	MSE=1.367e-05	nit=-1	nfev=40	t=9.00s
COBYLA	run 97/100	MSE=2.029e-05	nit=-1	nfev=59	t=13.18s
COBYLA	run 98/100	MSE=1.262e-03	nit=-1	nfev=400	t=92.11s
COBYLA	run 99/100	MSE=1.123e-05	nit=-1	nfev=109	t=27.49s
COBYLA	run 100/100	MSE=9.366e-02	nit=-1	nfev=20	t=4.53s

Scenario A - trust-constr (100 runs)

trust-constr	run 1/100	MSE=1.052e-04	nit=96	nfev=318	t=72.38s
trust-constr	run 2/100	MSE=1.055e-04	nit=32	nfev=78	t=15.89s
trust-constr	run 3/100	MSE=4.059e-05	nit=100	nfev=378	t=75.95s
trust-constr	run 4/100	MSE=4.197e-08	nit=100	nfev=291	t=58.78s
trust-constr	run 5/100	MSE=4.840e-08	nit=100	nfev=291	t=60.72s
trust-constr	run 6/100	MSE=1.912e-07	nit=100	nfev=294	t=65.03s
trust-constr	run 7/100	MSE=5.356e-07	nit=100	nfev=297	t=65.64s
trust-constr	run 8/100	MSE=4.429e+02	nit=100	nfev=357	t=77.92s
trust-constr	run 9/100	MSE=6.127e-11	nit=100	nfev=315	t=69.64s
trust-constr	run 10/100	MSE=1.467e-10	nit=100	nfev=315	t=71.12s
trust-constr	run 11/100	MSE=1.476e-09	nit=100	nfev=309	t=116.87s
trust-constr	run 12/100	MSE=1.964e-10	nit=100	nfev=324	t=117.00s
trust-constr	run 13/100	MSE=1.354e-08	nit=100	nfev=291	t=59.66s
trust-constr	run 14/100	MSE=1.503e-08	nit=100	nfev=288	t=74.67s
trust-constr	run 15/100	MSE=1.782e-06	nit=100	nfev=288	t=91.21s
trust-constr	run 16/100	MSE=1.054e-04	nit=100	nfev=327	t=99.16s
trust-constr	run 17/100	MSE=3.516e-10	nit=100	nfev=318	t=101.42s
trust-constr	run 18/100	MSE=1.052e-04	nit=89	nfev=291	t=70.85s
trust-constr	run 19/100	MSE=2.230e-06	nit=100	nfev=291	t=62.21s
trust-constr	run 20/100	MSE=2.993e-10	nit=100	nfev=318	t=72.52s
trust-constr	run 21/100	MSE=1.329e-06	nit=100	nfev=288	t=82.41s
trust-constr	run 22/100	MSE=1.250e-06	nit=100	nfev=279	t=81.97s
trust-constr	run 23/100	MSE=1.052e-04	nit=100	nfev=333	t=102.92s
trust-constr	run 24/100	MSE=5.941e-05	nit=100	nfev=363	t=88.40s
trust-constr	run 25/100	MSE=3.528e-11	nit=100	nfev=309	t=53.36s

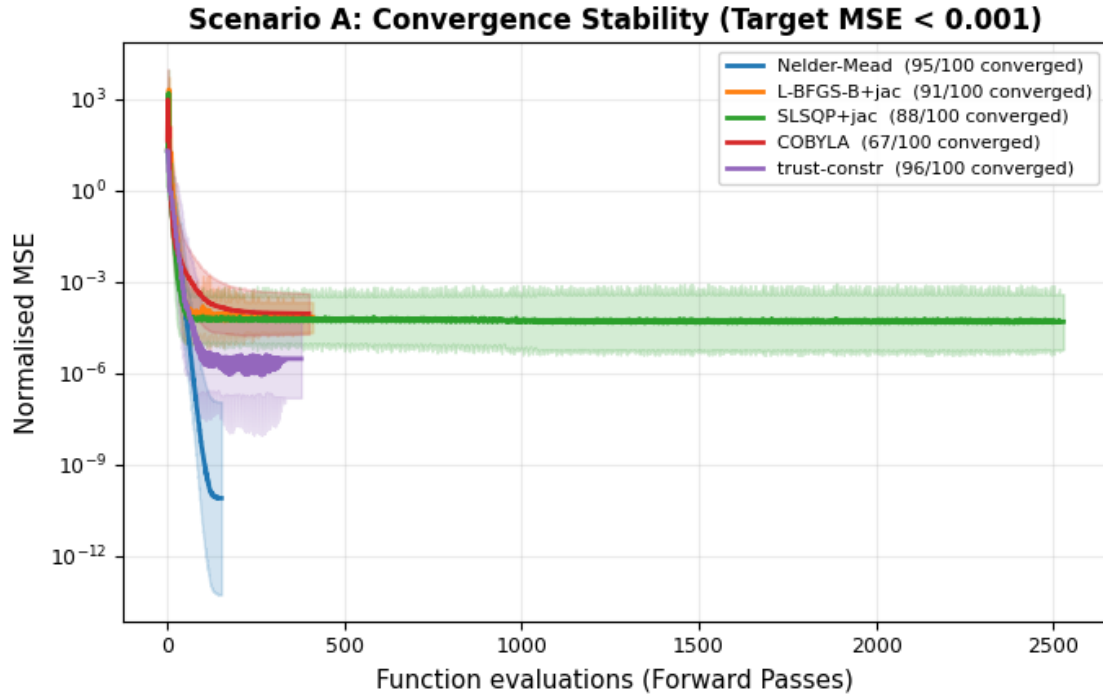
trust-constr	run 26/100	MSE=1.053e-04	nit=98	nfev=258	t=45.35s
trust-constr	run 27/100	MSE=9.326e+01	nit=100	nfev=375	t=67.21s
trust-constr	run 28/100	MSE=1.050e-04	nit=96	nfev=246	t=42.07s
trust-constr	run 29/100	MSE=9.242e-08	nit=100	nfev=291	t=49.90s
trust-constr	run 30/100	MSE=2.169e-06	nit=100	nfev=294	t=47.63s
trust-constr	run 31/100	MSE=1.052e-04	nit=94	nfev=309	t=48.50s
trust-constr	run 32/100	MSE=1.050e-04	nit=97	nfev=339	t=53.68s
trust-constr	run 33/100	MSE=1.027e-04	nit=100	nfev=360	t=57.04s
trust-constr	run 34/100	MSE=1.207e+03	nit=100	nfev=504	t=80.46s
trust-constr	run 35/100	MSE=4.943e-10	nit=100	nfev=291	t=46.06s
trust-constr	run 36/100	MSE=1.066e-06	nit=100	nfev=285	t=46.04s
trust-constr	run 37/100	MSE=2.168e-10	nit=100	nfev=327	t=51.44s
trust-constr	run 38/100	MSE=3.743e-10	nit=100	nfev=321	t=51.86s
trust-constr	run 39/100	MSE=2.191e-06	nit=100	nfev=288	t=45.62s
trust-constr	run 40/100	MSE=1.199e-08	nit=100	nfev=291	t=45.64s
trust-constr	run 41/100	MSE=1.161e-06	nit=100	nfev=285	t=44.76s
trust-constr	run 42/100	MSE=1.080e-06	nit=100	nfev=288	t=45.64s
trust-constr	run 43/100	MSE=3.139e-10	nit=100	nfev=318	t=55.29s
trust-constr	run 44/100	MSE=1.057e-08	nit=100	nfev=291	t=45.61s
trust-constr	run 45/100	MSE=1.054e-04	nit=97	nfev=315	t=50.01s
trust-constr	run 46/100	MSE=1.057e-04	nit=87	nfev=282	t=44.35s
trust-constr	run 47/100	MSE=1.062e-04	nit=47	nfev=129	t=20.38s
trust-constr	run 48/100	MSE=1.051e-04	nit=93	nfev=252	t=39.87s
trust-constr	run 49/100	MSE=2.204e-06	nit=97	nfev=279	t=43.50s
trust-constr	run 50/100	MSE=4.089e-08	nit=100	nfev=291	t=45.89s
trust-constr	run 51/100	MSE=9.030e-07	nit=100	nfev=273	t=43.03s
trust-constr	run 52/100	MSE=2.330e-06	nit=100	nfev=267	t=41.74s
trust-constr	run 53/100	MSE=1.485e-10	nit=100	nfev=333	t=52.03s
trust-constr	run 54/100	MSE=1.057e-04	nit=41	nfev=111	t=17.68s
trust-constr	run 55/100	MSE=3.856e-08	nit=100	nfev=297	t=47.08s
trust-constr	run 56/100	MSE=3.070e-10	nit=100	nfev=324	t=50.98s
trust-constr	run 57/100	MSE=1.054e-04	nit=36	nfev=90	t=14.54s
trust-constr	run 58/100	MSE=1.053e-04	nit=96	nfev=264	t=41.08s
trust-constr	run 59/100	MSE=1.002e-06	nit=100	nfev=297	t=46.71s
trust-constr	run 60/100	MSE=2.040e-10	nit=100	nfev=321	t=50.50s
trust-constr	run 61/100	MSE=1.589e-10	nit=100	nfev=333	t=52.15s
trust-constr	run 62/100	MSE=2.734e-10	nit=100	nfev=306	t=48.39s
trust-constr	run 63/100	MSE=1.052e-04	nit=96	nfev=237	t=37.08s
trust-constr	run 64/100	MSE=1.363e-06	nit=100	nfev=279	t=43.63s
trust-constr	run 65/100	MSE=1.333e-10	nit=100	nfev=321	t=50.83s
trust-constr	run 66/100	MSE=1.052e-04	nit=93	nfev=288	t=45.05s
trust-constr	run 67/100	MSE=1.273e-06	nit=100	nfev=294	t=46.05s
trust-constr	run 68/100	MSE=2.364e-06	nit=93	nfev=237	t=37.26s
trust-constr	run 69/100	MSE=9.153e-07	nit=100	nfev=294	t=46.08s
trust-constr	run 70/100	MSE=1.226e-08	nit=100	nfev=291	t=45.43s
trust-constr	run 71/100	MSE=1.053e-04	nit=100	nfev=339	t=53.73s
trust-constr	run 72/100	MSE=8.198e+02	nit=100	nfev=432	t=67.59s
trust-constr	run 73/100	MSE=6.662e-11	nit=100	nfev=312	t=49.29s

trust-constr	run 74/100	MSE=1.057e-04	nit=48	nfev=126	t=20.34s
trust-constr	run 75/100	MSE=4.383e-07	nit=100	nfev=291	t=46.00s
trust-constr	run 76/100	MSE=1.052e-04	nit=97	nfev=306	t=47.85s
trust-constr	run 77/100	MSE=1.090e-06	nit=100	nfev=291	t=46.25s
trust-constr	run 78/100	MSE=8.854e-07	nit=100	nfev=294	t=46.31s
trust-constr	run 79/100	MSE=1.054e-04	nit=100	nfev=342	t=53.97s
trust-constr	run 80/100	MSE=2.270e-09	nit=100	nfev=315	t=49.22s
trust-constr	run 81/100	MSE=1.054e-04	nit=100	nfev=327	t=51.80s
trust-constr	run 82/100	MSE=1.585e-08	nit=100	nfev=291	t=46.20s
trust-constr	run 83/100	MSE=1.056e-04	nit=99	nfev=255	t=39.81s
trust-constr	run 84/100	MSE=1.052e-04	nit=100	nfev=291	t=45.75s
trust-constr	run 85/100	MSE=1.406e-07	nit=100	nfev=291	t=45.38s
trust-constr	run 86/100	MSE=1.056e-04	nit=100	nfev=333	t=52.47s
trust-constr	run 87/100	MSE=1.793e-09	nit=100	nfev=294	t=46.17s
trust-constr	run 88/100	MSE=8.593e-11	nit=100	nfev=312	t=49.05s
trust-constr	run 89/100	MSE=4.525e-07	nit=100	nfev=291	t=45.88s
trust-constr	run 90/100	MSE=1.053e-04	nit=93	nfev=186	t=29.22s
trust-constr	run 91/100	MSE=3.244e-10	nit=100	nfev=318	t=50.32s
trust-constr	run 92/100	MSE=2.171e-06	nit=100	nfev=276	t=43.55s
trust-constr	run 93/100	MSE=1.052e-04	nit=86	nfev=273	t=42.52s
trust-constr	run 94/100	MSE=4.467e-08	nit=100	nfev=297	t=47.40s
trust-constr	run 95/100	MSE=2.711e-08	nit=100	nfev=294	t=46.88s
trust-constr	run 96/100	MSE=1.051e-04	nit=98	nfev=357	t=55.43s
trust-constr	run 97/100	MSE=1.053e-04	nit=100	nfev=276	t=42.93s
trust-constr	run 98/100	MSE=1.048e-04	nit=96	nfev=333	t=52.24s
trust-constr	run 99/100	MSE=1.057e-04	nit=29	nfev=69	t=11.28s
trust-constr	run 100/100	MSE=5.034e-07	nit=100	nfev=291	t=46.09s

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SCENARIO A: Robustness (Tol < 0.001) & Discovered Physics State vs Reference
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method_tag  conv_rate nit_mean nfev_mean wall_time_mean final_mse_mean I_1_ref
I_1      I_3_ref   I_3      alpha_x   alpha_y
COBYLA      67%      184.4      184.4      57.646 s      8.21e-05      0.8218 A
0.7429 A 1.0430 A 0.5804 A 5.41e-04 -2.58e-03
L-BFGS-B+jac 91%      12.7      166.7      32.301 s      4.49e-05      0.8218 A
0.7429 A 1.0430 A 0.5805 A 1.96e-03 4.64e-03
Nelder-Mead 95%      57.7      113.8      22.563 s      5.76e-11      0.8218 A
0.2958 A 1.0430 A 0.2830 A -7.73e-08 -6.31e-07
SLSQP+jac   88%      35.0      357.3      61.763 s      4.54e-05      0.8218 A
0.7432 A 1.0430 A 0.5807 A -1.79e-04 -1.21e-05
trust-constr 96%      95.2      292.2      53.498 s      4.92e-07      0.8218 A
0.2958 A 1.0430 A 0.2830 A -1.76e-04 3.36e-04
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Scenario A: Best Optimised Quadrupole Currents per Method

Method	Best MSE	I (A)	I2 (A)
Nelder-Mead	2.7877e-14	0.2958	0.2830
L-BFGS-B+jac	1.2465e-06	0.7429	0.5805
SLSQP+jac	1.6069e-08	0.7432	0.5807
COBYLA	1.5288e-06	0.7429	0.5804
trust-constr	3.5285e-11	0.2958	0.2830

Reference FELsim S1 currents: I=0.8218 A, I2=1.0430 A

Overall best across all methods/runs:

Method: Nelder-Mead, MSE=2.7877e-14

I=0.29581 A, I2=0.28299 A

time: 6h 20min 26s (started: 2026-06-02 16:39:08 -10:00)